Intent-phrase planning: proposal helps, reranking hurts

TextJEPA research cycle

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The concrete question

- The model chooses a natural-language instruction, and the environment executes it.
- A supervised policy prior proposes likely next instructions.
- JEPA predicts latent consequences and reranks those proposals.
- Success means solving within the minimum action count, or with two extra actions.

The prior is stronger than current JEPA reranking

Actions	JEPA strict	Prior strict	JEPA +2	Prior +2
3	.172	**.361**	.511	**.744**
6	.133	**.178**	.417	**.661**
9	.067	**.183**	.744	**.872**

- What to notice: every nonzero in-range cell favors prior-only planning.
- Why it matters: current latent energy removes value rather than adding it.

Deep rollout helps only with privileged future menus

- At length nine, strict top-four success rises from .061 at depth one to .439 at depth four.
- With two extra actions, the same comparison rises from .750 to .994.
- Future feasible actions came from the hidden dependency graph, so this is an oracle diagnostic.
- Wide hierarchical CEM loses to the first-feasible rule at every tested bottleneck.

Next decision: separate two kinds of difficulty

- The old curve changed reasoning length and prompt size at the same boundary.
- We will vary prompt variables while holding reasoning length at nine.
- We will vary reasoning length from nine to twelve while holding prompts at 24–36 variables.
- Only then will we train a deployable predicted-state future-action proposal model.